Getting the Split-Plot Right

Mixed model analysis of an oat variety × nitrogen field trial in R | Case study | Self-initiated portfolio project using public data

The problem

Split-plot experiments are common in agronomy: one factor (here, variety) is applied to large whole plots, and a second factor (nitrogen) to smaller subplots within them. Because the two factors are randomized at different scales, they need different error terms. A frequent mistake is to analyse the trial as an ordinary randomized complete block design (RCBD), which tests every effect against one pooled error. This case study shows the correct analysis and what goes wrong with the incorrect one.

Data

Item	Detail
Source	Yates (1935) oats experiment, public dataset oats in the R package MASS
Design	6 blocks; variety on whole plots (3 levels); nitrogen on subplots (0, 0.2, 0.4, 0.6 cwt/acre)
Size	72 subplots = 6 blocks × 3 varieties × 4 nitrogen rates
Response	Grain yield (¼ lb per subplot, original units)

Methods

- **Incorrect analysis (for comparison):** two-way ANOVA with blocks, as if the trial were an RCBD.
- **Correct analysis:** split-plot ANOVA with two error strata, and the equivalent linear mixed model Yield ~ Variety × Nitrogen + (1 | Block) + (1 | Block:Variety), fitted by REML (lme4) with Kenward-Roger F tests (lmerTest).
- **Assumption checks:** residual plots, Shapiro-Wilk, Levene's test, and a Q-Q plot of whole-plot effects.
- **Comparisons:** estimated marginal means (emmeans), Tukey-adjusted letters, and polynomial contrasts for the quantitative nitrogen factor.

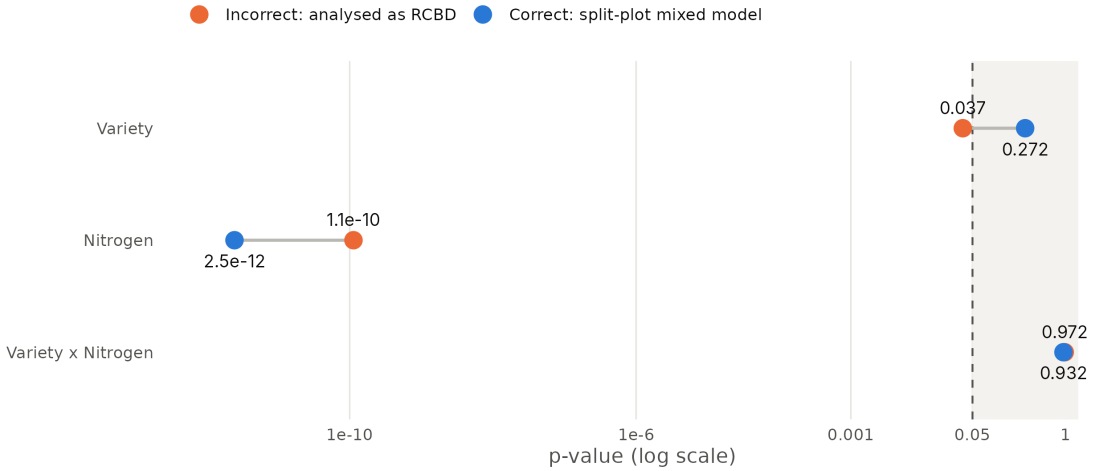
Result 1: the incorrect analysis changes the conclusion

Effect	Incorrect RCBD analysis	Correct split-plot mixed model
Variety	F = 3.51, p = 0.037 (error df 55)	**F = 1.49, p = 0.27** (error df 10)
Nitrogen	F = 26.25, p < 0.001	F = 37.69, p < 0.001
Variety × Nitrogen	F = 0.21, p = 0.97	F = 0.30, p = 0.93

Treated as an RCBD, varieties appear to differ significantly. In the correct analysis, variety is tested against the whole-plot error (Block × Variety), which is larger and has only 10 degrees of freedom, and the difference is not significant. A report based on the incorrect analysis would recommend a variety the data do not support.

The wrong (RCBD) analysis makes the variety effect falsely significant

p-values for the same data under both analyses. Shaded area: p > 0.05 (not significant)

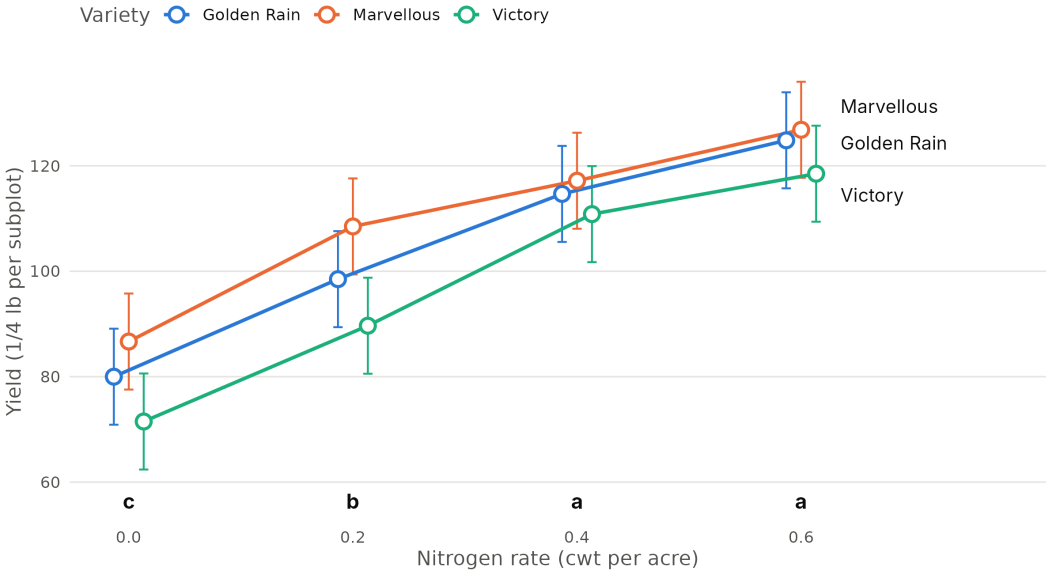


The RCBD analysis tests Variety against one pooled error (55 df) instead of the whole-plot error (Block x Variety, 10 df).

Result 2: nitrogen response

Oat yield responds to nitrogen; varieties do not differ significantly

Means +/- SE from the split-plot mixed model. Letters: nitrogen groups (Tukey, p < 0.05)



Data: Yates (1935) oats split-plot trial, R package MASS. Self-initiated portfolio project using public data.

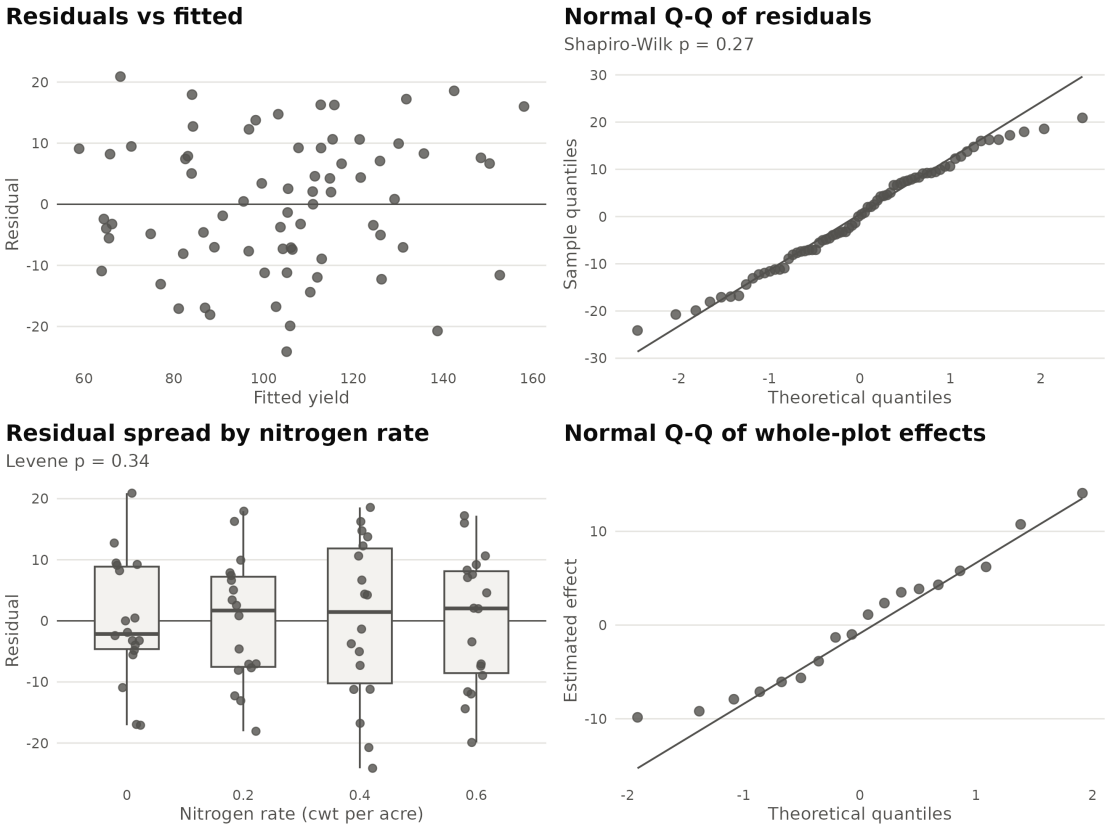
- Nitrogen strongly increased yield (p < 0.001), from 79.4 at 0 cwt to 123.4 at 0.6 cwt per acre.
- The response is mainly linear (linear contrast p < 0.001; quadratic p = 0.11). Yields at 0.4 and 0.6 cwt did not differ significantly (Tukey).
- No variety × nitrogen interaction (p = 0.93): the three varieties responded to nitrogen in a similar way.

Nitrogen (cwt/acre)	Mean yield	SE	95% CI	Tukey group
0	79.4	7.17	55.3 – 103.5	c
0.2	98.9	7.17	74.8 – 123.0	b
0.4	114.2	7.17	90.2 – 138.3	a
0.6	123.4	7.17	99.3 – 147.5	a

Confidence intervals include block-to-block variation, while Tukey comparisons use the smaller within-block error, so overlapping intervals can still differ significantly.

Model checks

Model diagnostics: no strong evidence against normality or equal variance



- No strong evidence against normality (Shapiro-Wilk p = 0.27) or equal variance across nitrogen rates (Levene p = 0.34). Residual spread across varieties was borderline (Levene p = 0.06) and is reported.
- Variance components: blocks 43%, whole-plot error 21%, subplot error 36% of total variance. The two error levels are clearly different, which is why the split-plot structure matters.

Limitations

This is a single historical trial with six blocks, so the whole-plot test of variety has only 10 degrees of freedom and limited power: a non-significant result does not prove that the varieties are equal. Yield is reported in the dataset's original units.

What a client receives in this kind of project

- A fully commented, reproducible R script (data check, models, assumption tests, comparisons, figures).
- ANOVA and mean-comparison tables with letter groupings, ready for a manuscript or report.
- Publication-ready figures at 300 dpi.
- A short methods and results note, including the limitations of the analysis.

Software: R 4.3.3 with lme4, lmerTest, pbkrtest, emmeans, multcomp, car, ggplot2 and patchwork. References: Yates F. (1935) Complex experiments. Suppl. J. R. Stat. Soc. 2:181–247; Venables W.N. & Ripley B.D. (2002) Modern Applied Statistics with S, 4th ed. Springer.